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Unit 3 — Algorithms and Problem Solving

A CS50-Style Deep Dive Study Guide for 11th Class Computer Science

Welcome, Future Algorithmic Thinker

Here's a secret: you already solve problems algorithmically every single day. When you decide the fastest route to school, when you search for a contact in your phone, when you sort your playlist by favorite songs — you're running an algorithm in your head, whether you realize it or not.

This unit gives that instinct a name, a structure, and a set of powerful tools. We're going to learn:

- What makes a problem a **computational problem**, and how to describe one clearly
- A brute-force strategy called **Generate-and-Test** that can crack almost any small problem
- The difference between problems that **can** be solved and problems that **provably cannot**
- How to measure whether a solvable problem is *fast* to solve or *painfully slow* — using **Big O notation**
- Four powerful **algorithm design techniques**: Divide and Conquer, Greedy, Dynamic Programming, and Backtracking
- The algorithms every computer scientist knows by heart: **Bubble Sort, Selection Sort, Linear Search, Binary Search, BFS, and DFS**

One thing before we start: **confusion is completely normal here**. Concepts like Big O notation and P vs. NP confuse even senior computer science students and professional researchers.

Nobody expects you to master this in one read. Our job is to build your intuition, one small idea at a time, until the formal notation stops looking scary and starts looking like a helpful shortcut.

Let's start thinking like algorithm designers.

Introduction

Understanding algorithms is essential — not only for computer science, but for everyday problem-solving. In this unit, we'll start by learning what **computational problems** are and how to describe them clearly. Then we'll explore different types of algorithms and how they help us solve various kinds of problems. Finally, we'll learn how to **measure the efficiency** of algorithms, so we can choose the best solution for any given task.

Student Learning Outcomes

By the end of this chapter, you will be able to:

- Describe and categorize different types of computational problems
- Explain why algorithms matter for problem-solving
- Apply the Generate-and-Test method to solve computational problems
- Differentiate between solvable and unsolvable problems
- Understand problem complexity and categorize problems into P, NP, NP-Hard, and NP-Complete
- Identify common computational problems like sorting and searching
- Apply algorithm design techniques: divide and conquer, greedy methods, and dynamic programming
- Implement and compare Bubble Sort, Binary Search, BFS, and DFS
- Evaluate algorithms in terms of efficiency and scalability
- Develop algorithmic thinking to solve problems systematically

3.1 Understanding Computational Problems

The Hook (Story Mode)

Imagine handing someone a recipe that just says "bake until delicious." How long? At what temperature? Delicious according to whom? That recipe is useless — not because the cook is bad, but because the *instructions* are broken. Now imagine a recipe that says "bake at 180°C for exactly 20 minutes." Suddenly, anyone — even a computer — could follow it.

Computers are exactly like that second cook: powerful, but only if you give them a problem described with total clarity. That's what this section is about.

The Explanation

A **computational problem** is a challenge that can be solved through a computational process — meaning it can be solved using an **algorithm**: a set of step-by-step instructions that a computer can execute.

Every computational problem has three essential parts:

- **Input** — the data or information given to the algorithm at the beginning.
- **Process** — the steps or rules (the algorithm itself) applied to the input to generate the output.
- **Output** — the solution or result produced after processing the input.

ANATOMY OF A COMPUTATIONAL PROBLEM

[INPUT] ----> [PROCESS (the algorithm)] ----> [OUTPUT]

e.g. "5"

"multiply by 2"

"10"

Quick Recap

Every computational problem boils down to the same three ingredients: input, process, and output — and an algorithm is simply the recipe that connects them.

3.1.1 Characterizing Computational Problems

The Explanation

To solve a problem computationally, we first need to understand its *characteristics*. This means identifying:

- The **inputs** the problem starts with
- The **desired outputs** we want to end up with
- The **process** needed to transform inputs into outputs

Skipping this step is like trying to build a house without first knowing what rooms the owner wants — you might build something technically impressive that solves the wrong problem entirely.

Quick Recap

Before writing any algorithm, always identify exactly what goes in, what should come out, and what transformation connects them.

3.1.1.1 Classifying Computational Problems

The Hook (Story Mode)

Think about the different kinds of questions you ask a search engine: "Is this website safe?" (yes/no), "Find me the fastest route to Lahore" (a specific answer), "What's the cheapest flight to Karachi?" (the *best* answer among many), "How many ways can I split this bill among friends?" (a count). Each of these is a genuinely different *type* of problem, and computer scientists have names for each type.

The Explanation

Computational problems can be classified into different categories based on their characteristics and the methods required to solve them:

- **Decision Problems** — the output is a simple "yes" or "no." *Example: "Is this number even?"*
- **Search Problems** — the task is to find a solution or item that meets certain criteria. *Example: "Find a book titled 'Data Structures' in a library catalog."*
- **Optimization Problems** — the goal is to find the *best* solution according to some criteria. *Example: "Find the cheapest flight from Lahore to Dubai."*
- **Counting Problems** — the objective is to count the number of ways certain conditions can be met. *Example: "How many different ways can you make change for 100 rupees?"*

Category	Question It Answers	Example
Decision	Yes or no?	Is this number prime?
Search	Where/what is it?	Find "Islamabad" in a list of cities
Optimization	What's the best option?	Shortest delivery route
Counting	How many ways?	Number of possible passwords

Interactive Stop-Point

Pause & Think: Classify each of these as Decision, Search, Optimization, or Counting: (1) "Is this password strong enough?" (2) "What is the shortest path from home to school?" (3) "How many unique usernames can be made using 6 letters?"

Quick Recap

Not all computational problems ask the same kind of question — recognizing whether a problem wants yes/no, a specific item, the *best* item, or a count changes how you'll approach solving it.

3.1.1.2 Well-defined vs. Ill-defined Problems

The Hook (Story Mode)

Remember our recipe example? "Bake until delicious" is an **ill-defined** problem — vague, subjective, with no clear stopping point. "Bake at 180°C for exactly 20 minutes" is a **well-defined** problem — precise, measurable, and something a machine (or a person) could follow without guessing.

The Explanation

Problems can also be categorized by how clearly they are defined:

Well-defined Problems have clear goals, inputs, processes, and outputs. *Example: "Determine if a number is even."* This is well-defined because it has:

- A clear **goal**: determine if the number is even
- A clear **input**: a single integer
- A clear **process**: check if the number is divisible by 2
- A clear **output**: "Even" or "Odd"

Ill-defined Problems lack clear definitions, or have ambiguous goals and requirements.

Example: "How to reduce poverty in Pakistan." This goal is vague and broad — there's no single clear input, no agreed-upon process, and no measurable output that everyone would accept as "done."

FLOWCHART: "IS THE NUMBER EVEN?" (a well-defined problem)

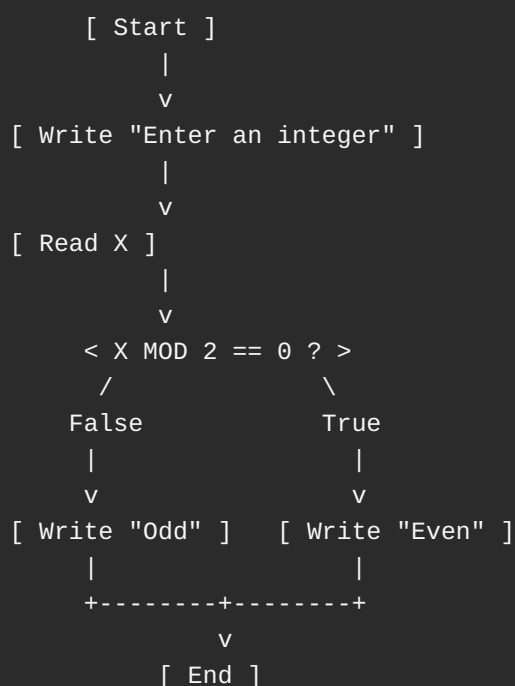


Figure 3.1: Finding an Even Number

The Practical Walkthrough — Testing Whether a Problem Is Well-Defined

1. **State the goal in one sentence.** Can you write it without using vague words like "good," "best," "nice," or "better"? If not, it's a warning sign.
2. **Identify the exact input.** Can you name its data type (a number? a list? text?) precisely?
3. **Identify the exact process.** Could you write down the steps as a numbered list that never requires personal judgment?

4. **Identify the exact output.** Would two different people, given the same input and following the same steps, always produce the *same* output?
5. **If all four checks pass** — goal, input, process, output — the problem is well-defined. If any check fails, it's ill-defined, and you likely need to *narrow* the problem before an algorithm can solve it.

What just happened? You just built a personal checklist for spotting the difference between a solvable computational problem and a vague real-world goal that needs to be broken down further before a computer can touch it.

Interactive Stop-Point

Pause & Think: Is "Design an algorithm to find the most beautiful picture on Instagram" a well-defined or ill-defined computational problem? Explain why, using the four-part checklist above.

Quick Recap

A well-defined problem gives a computer everything it needs — clear input, process, and output — while an ill-defined problem must first be narrowed down before any algorithm can attempt it.

3.2 Algorithms for Problem Solving

The Hook (Story Mode)

An algorithm is just a recipe — step-by-step instructions for turning ingredients (input) into a finished dish (output). You already follow algorithms constantly: how you tie your shoelaces, how you make tea, how you get ready for school in the morning. Computer science simply asks us to write these recipes with total precision, so a machine with zero common sense can follow them perfectly.

The Explanation

Algorithms are step-by-step procedures for solving problems, much like a recipe provides steps for cooking a dish. Understanding algorithms is essential because they provide the logic behind software operations, allowing us to solve complex problems, optimize performance, and ensure accuracy in various applications.

Did You Know? The Google search engine uses a complex algorithm called **PageRank** to determine the relevance of web pages. This algorithm considers factors like the number of links to a page and the quality of those links, to rank pages in search results.

Tidbit: When learning about algorithms, try to relate them to real-life tasks you already know. This will help you understand how algorithms work and why they matter.

Interactive Stop-Point

Pause & Think: Write out, step by step, the "algorithm" you personally follow every morning to get ready for school. Is every step precise enough that someone else could follow it exactly?

Quick Recap

An algorithm is simply a precise, repeatable recipe for solving a problem — and you're already surrounded by them, both in daily life and inside every app you use.

3.2.1 Generate-and-Test Method

The Hook (Story Mode)

Picture a 4-digit bike lock you've forgotten the combination to. With no other clues, your only option is to try 0000, then 0001, then 0002... checking each combination until one finally clicks open. That, exactly, is the **Generate-and-Test method** — computer science's formal name for "try everything until something works."

The Explanation

The **Generate-and-Test method** works by **generating** potential solutions to a problem, and then **testing** each one to determine if it meets the required conditions. The process continues until a satisfactory solution is found, or all possible solutions have been exhausted.

This method is particularly useful when:

- The **problem space is small**, making it feasible to generate and test all possible solutions.
- There is **no clear strategy** for finding a solution, so an exhaustive search is necessary.
- **Heuristics** (rules of thumb that narrow the search) can be applied to reduce the number of generated solutions, making the process more efficient.

GENERATE-AND-TEST FLOWCHART

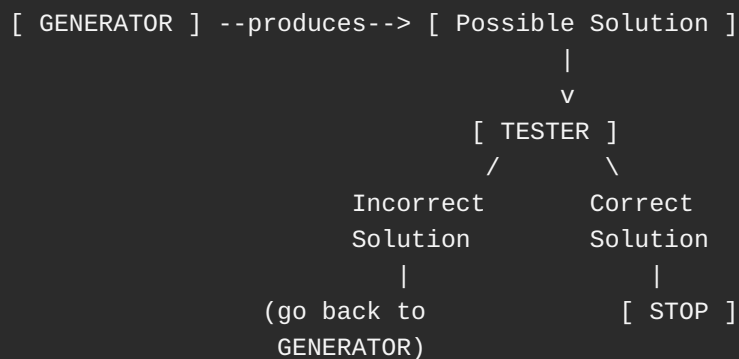


Figure 3.2: Flowchart of the Generate-and-Test Method

Did You Know? The Generate-and-Test method is often used in AI applications, such as game playing and problem-solving, where the solution space is large, and the best approach is to try different possibilities until one works!

The Practical Walkthrough — Cracking a 2-Digit Lock with Generate-and-Test

1. **Define the problem space.** A 2-digit lock has 100 possible combinations: 00 through 99.
2. **Generate the first candidate.** Try **00**.
3. **Test it.** Does the lock open? No.
4. **Generate the next candidate.** Try **01**.
5. **Test it.** No.
6. **Repeat** this generate-test cycle, incrementing by 1 each time, until a candidate opens the lock — say, at **47**.
7. **Stop.** The correct solution has been found after 48 total attempts (00 through 47).

What just happened? You just traced the entire Generate-and-Test loop by hand — it's simple, guaranteed to eventually find the answer (if one exists), but it can be slow when the problem space is large.

Interactive Stop-Point

Grab a Partner: Partner A picks a secret whole number between 1 and 20 (without telling Partner B). Partner B uses Generate-and-Test — guessing 1, then 2, then 3, and so on — until they find it. Count the total guesses it took. Keep this number in mind; we'll compare it to a smarter method later in this unit.

Quick Recap

Generate-and-Test is the brute-force strategy of trying every possible solution until one works — reliable, but potentially slow when there are many possibilities to check.

3.3 Problem Solvability and Complexity

The Hook (Story Mode)

In 1936 — before electronic computers even existed — a young mathematician named **Alan Turing** proved something astonishing: there exist problems that **no algorithm, running on any computer, no matter how powerful, will ever be able to solve for every possible case**. This wasn't a limitation of 1930s technology; it's a permanent, mathematical fact about the nature of computation itself. Understanding *which* problems can be solved — and how efficiently — is exactly what this section is about.

The Explanation

Problem solvability and complexity help us determine **whether** a problem can be solved using an algorithm, and **if so, how efficiently** it can be solved.

Quick Recap

Before asking "how do I solve this efficiently?" computer scientists first ask "can this even be solved at all?" — and surprisingly, the answer is sometimes no.

3.3.1 Solvable vs. Unsolvable Problems

The Explanation

In computer science, problems are classified as **solvable** or **unsolvable**, based on whether there exists an algorithm that can provide a solution.

Solvable Problems — a problem is solvable if an algorithm can solve it within a **finite amount of time**. These problems have clearly defined inputs and outputs, and there's a step-by-step procedure to reach the solution.

Example: Calculating the **Greatest Common Divisor (GCD)** of two integers is a solvable problem. The **Euclidean algorithm** provides a clear, finite method to determine the GCD, making it a classic example of a solvable problem.

Unsolvable Problems — a problem is unsolvable if **no algorithm can be created** that will provide a solution in *all* cases. These problems have no general procedure that guarantees a solution for every possible input.

Example: The **Halting Problem** is the most famous unsolvable problem. It asks: given a program and its input, will the program eventually stop running (halt), or will it run forever? **Alan Turing proved in 1936** that no general algorithm can solve the Halting Problem for all possible program-input pairs — making it a foundational example of an unsolvable problem.

Tidbit: When tackling complex problems, it's essential to first determine whether the problem is solvable at all. This saves time and resources by ensuring you're working on a problem that *can* actually be resolved using an algorithm.

Interactive Stop-Point

Pause & Think: If you wrote a program that tries to predict whether *any* other program will eventually crash or loop forever, why might this be an impossible task to solve perfectly for every single program that could ever be written?

Quick Recap

Solvable problems have a guaranteed algorithm that finishes in finite time; unsolvable problems — like Turing's Halting Problem — provably have no algorithm that works for every case.

3.3.2 Tractable vs. Intractable Problems

The Hook (Story Mode)

Imagine two solvable problems: sorting your 30 classmates by height (easy, fast, done in seconds) versus finding the absolute shortest road trip that visits every single city in Pakistan exactly once (technically solvable, but for a large number of cities, no computer on Earth could finish it in your lifetime). Both are "solvable" — but their *practical* difficulty is worlds apart.

The Explanation

Once a problem is confirmed solvable, the next question is its **computational complexity** — how *efficiently* it can be solved. Problems are categorized as **tractable** or **intractable** based on the time and space resources required to solve them.

Tractable Problems — solvable in **polynomial time**, denoted **P**. Polynomial time means the time taken grows at a *manageable* rate (as a polynomial function) relative to the input size.

Tractable problems are considered "efficiently solvable."

Example: Sorting a list of numbers using **Merge Sort** or **Quick Sort** is tractable, with a time complexity of $O(n \log n)$, where n is the number of elements.

Intractable Problems — require **super-polynomial time**, often growing exponentially with input size. These problems become impractical to solve for large inputs, because the time required becomes unmanageable.

Example: The **Traveling Salesman Problem (TSP)** — finding the shortest route that visits a set of cities and returns to the start — is intractable. It's **NP-hard**, and as the number of cities increases, the number of possible routes grows *factorially*, making it infeasible to solve exactly for large instances.

Interactive Stop-Point

Pause & Think: If a tractable algorithm takes 1 second to process 10 items, and doubling the input roughly doubles the time (a rough intuition for polynomial growth), estimate how long it might take to process 1,000 items. Now imagine an intractable algorithm where doubling the input *squares* the time instead — how quickly does that spiral out of control?

Quick Recap

Tractable problems scale up at a manageable, polynomial rate; intractable problems scale up explosively, quickly becoming impossible to solve exactly as the input grows.

3.3.3 Complexity Classes (P, NP, NP-hard, NP-complete)

The Hook (Story Mode)

In 1971 and 1973, two computer scientists — **Stephen Cook** and, independently, **Leonid Levin** — formulated what's now called the **P vs. NP question**, one of the most important unsolved problems in all of mathematics and computer science. It's so important that the Clay Mathematics Institute now offers a **\$1,000,000 Millennium Prize** to anyone who can definitively prove whether P equals NP or not. Nobody has claimed it yet.

The Explanation

Understanding the complexity of problems involves classifying them into categories based on their solvability and the time required to solve them.

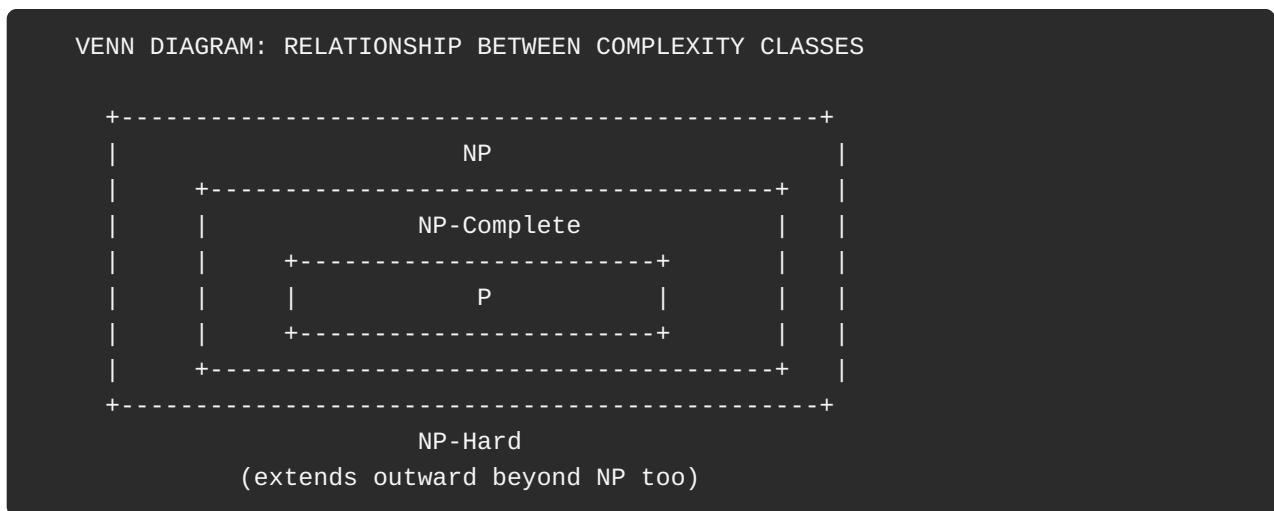


Figure 3.4: Venn diagram of the complexity classes P , NP , NP -hard, and NP -complete

Did You Know? The question of whether **P equals NP** is one of the most important unsolved problems in computer science. It has major implications for cryptography, algorithm design, and our overall understanding of computational complexity.

Quick Recap

P , NP , NP -hard, and NP -complete are nested categories that describe *how* easy or hard problems are to solve and to verify — and whether P actually equals NP remains one of the biggest open mysteries in all of computing.

3.3.3.1 Class P

The Explanation

Class P refers to problems that can be solved **efficiently** by a computer — meaning a computer can find a solution quickly, even as the problem size grows.

Example: Sorting a list of numbers, like `[4, 1, 3, 2, 5]`, into ascending order

`[1, 2, 3, 4, 5]`. The time required to sort the list grows at a manageable rate as the list size increases — going from 5 numbers to 10 numbers increases the time, but it stays within a reasonable limit.

Interactive Stop-Point

Pause & Think: Name two everyday computer tasks (searching your phone's contacts, sorting emails by date, etc.) that you'd expect to be in Class P — fast even as the amount of data grows.

Quick Recap

Class P problems are the "easy" problems of computer science — solvable quickly and predictably, even at large scale.

3.3.3.2 Class NP

The Explanation

Class NP refers to problems for which, **if a solution is given, it can be checked quickly** by a computer. These are problems where *verifying* a proposed solution is easy, even if *finding* that solution in the first place might be difficult and time-consuming.

Example: Solving a **Sudoku** puzzle can be very hard — but if someone hands you a *completed* Sudoku grid, checking whether every row, column, and 3×3 sub-grid contains the digits 1–9 exactly once is quick and easy.

Figure 3.3: A simple Sudoku Puzzle — hard to solve, easy to check once solved.

Interactive Stop-Point

Pause & Think: A friend hands you a completed jigsaw puzzle and claims they solved it correctly. Is checking their answer easier or harder than solving the puzzle yourself from scratch? How does this relate to the definition of Class NP?

Quick Recap

Class NP problems may be hard to *solve*, but any proposed solution can be *verified* quickly — that's the defining trait of this class.

3.3.3.3 Class NP-Hard

The Explanation

NP-hard problems are **at least as difficult as the hardest problems in NP**. Solving an NP-hard problem is extremely challenging, and no efficient (polynomial-time) algorithm is currently known for finding a solution.

Example: The **Traveling Salesman Problem (TSP)**, discussed in section 3.3.2, is a well-known NP-hard problem.

Quick Recap

NP-hard problems sit at or beyond the outer edge of computational difficulty — they're at least as tough as anything else in NP, and often tougher.

3.3.3.4 NP-Complete

The Explanation

NP-Complete problems form a special subset of NP problems: they are **both in NP and as hard as the hardest problems in NP**. This means these problems are particularly challenging — and remarkably, if you could solve *just one* NP-Complete problem efficiently, you could solve *all* NP problems efficiently.

Example: The **Knapsack Problem** is a classic NP-Complete problem. You have a knapsack with a maximum weight capacity, and a set of items, each with a weight and a value. The goal is to determine the most valuable combination of items that fits without exceeding the weight capacity.

Interactive Stop-Point

Pause & Think: If someone proves that P equals NP — meaning every problem whose solution can be *checked* quickly could also be *solved* quickly — how might that instantly impact global website security and bank encryption systems, which often rely on certain problems being hard to solve?

Quick Recap

NP-Complete problems are the "hardest of the hard" within NP — solving any single one efficiently would unlock efficient solutions to every problem in NP, which is exactly why the P vs. NP question is such a massive deal.

3.4 Algorithm Analysis

The Hook (Story Mode)

Imagine two delivery drivers given the same 100 packages. One methodically plans an efficient route; the other just drives randomly and hopes for the best. Both might eventually deliver everything — but one will take far longer, waste far more fuel, and scale terribly as the number of packages grows. **Algorithm analysis** is how computer scientists tell these two drivers apart *before* either of them starts driving.

The Explanation

Algorithm analysis is the process of determining the computational complexity of an algorithm — including its **time complexity** and **space complexity**. This analysis helps predict an algorithm's performance and is crucial for selecting the best algorithm for a given task.

Quick Recap

Algorithm analysis lets us predict how "good" an algorithm really is *before* we ever run it on a huge, real-world dataset.

3.4.1 Time Complexity

The Explanation

Time complexity measures how the *running time* of an algorithm increases as the size of the input data grows. It helps us understand how efficiently an algorithm performs as it handles larger amounts of data.

Example: Sorting a short list of numbers is quick. As the list grows longer, the time required to sort it also grows. Time complexity lets us predict exactly *how* this runtime scales with input size.

Quick Recap

Time complexity isn't about how fast an algorithm runs on *one* input — it's about how its speed changes as the input keeps growing.

3.4.1.1 Big O Notation

The Hook (Story Mode)

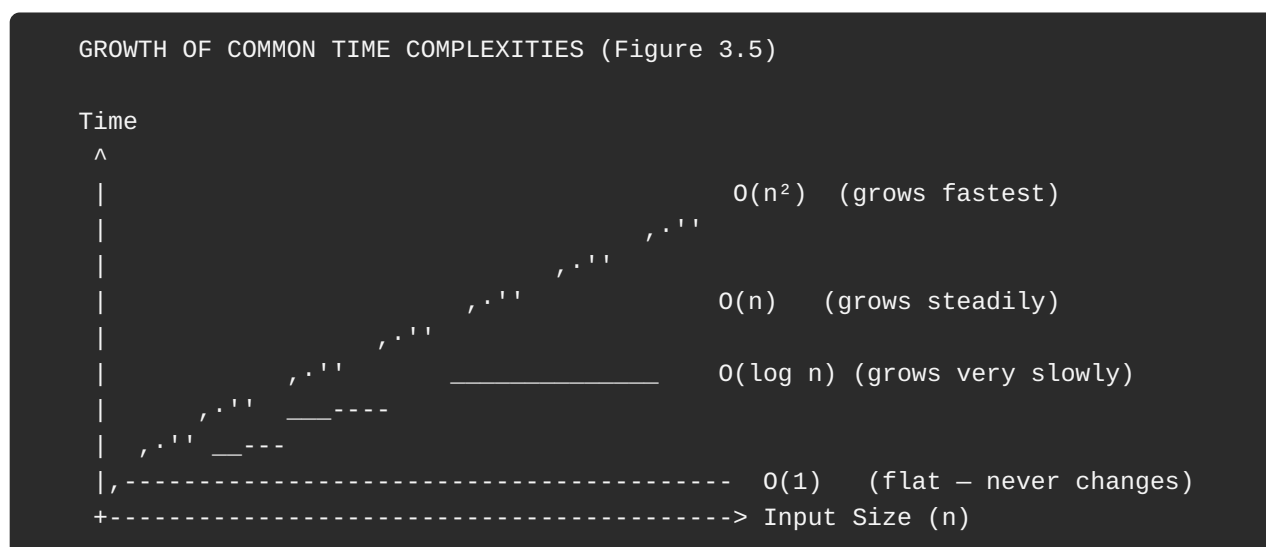
Long before computers existed, telephone directory operators could find any name among *millions* of alphabetically sorted entries in under 20 flips of the page — not by checking every name, but by repeatedly cutting the search area in half. That instinct — "cut the problem down instead of checking everything" — is precisely what **Big O notation** helps us measure and compare.

The Explanation

Big O notation is a mathematical shorthand for describing an algorithm's time complexity. Think of it as a friendly shortcut for counting steps as input sizes grow huge — not an intimidating college-level formula. It gives an **upper bound** on the time an algorithm will take as input size grows, making it easy to compare the efficiency of different algorithms.

How Big O Notation Works — Common Examples:

- **$O(1)$ — Constant Time:** The runtime stays the same, no matter the input size. *Example: checking if the first item in a list exists.*
- **$O(n)$ — Linear Time:** The runtime grows linearly with input size. *Example: searching through a list of n students one by one to find a specific student ID — the time depends directly on n .*
- **$O(n^2)$ — Quadratic Time:** The runtime grows with the square of the input size. *Example: comparing every pair of n students in a programming competition to determine the best team — the number of comparisons is roughly $n(n-1)$, which is quadratic growth.*
- **$O(\log n)$ — Logarithmic Time:** The runtime grows very slowly relative to input size. *Example: guessing a number between 1 and 100 using only yes/no questions like "Is it greater than 50?" — each question cuts the range in half, which is exactly how Binary Search works.*



When comparing complexities: **$O(1)$** stays completely flat no matter how large n gets. **$O(\log n)$** rises very slowly. **$O(n)$** rises steadily, in a straight line. **$O(n^2)$** rises the fastest of the four, curving sharply upward.

Did You Know? Big O notation helps computer scientists understand an algorithm's efficiency in the **worst-case scenario**, allowing them to predict how well it will perform as input data grows.

The Practical Walkthrough — Classifying an Algorithm's Big O by Counting Steps

1. Write out the algorithm's steps in plain pseudocode.

```
for each student in list of n students:  
    check if student.id == target_id
```

2. Ask: does the number of operations depend on n at all? Here, yes — we loop through every student.
3. Ask: does the loop run once through the list, or does it contain a nested loop over the list again? Here, it's a single loop — one pass through n students.
4. Classify accordingly. A single pass through n items → $O(n)$, Linear Time.
5. Compare with a nested-loop version (comparing every student against every other student) → two nested loops over n items → $O(n^2)$, Quadratic Time.

What just happened? You just learned the single most useful trick in this entire unit: **count your loops**. One loop over the input is usually $O(n)$. A loop inside a loop is usually $O(n^2)$. Repeatedly cutting the input in half is $O(\log n)$.

Interactive Stop-Point

Grab a Partner: Partner A picks a secret whole number between 1 and 100. Partner B guesses using **Binary Search** — always asking "Is it higher or lower than X ?" and cutting the range in half each time. Count the guesses. Then try guessing the *same* number using **Linear Search** — guessing 1, then 2, then 3, in order. Compare how many guesses each method took, and connect that back to your earlier bike-lock guessing exercise from section 3.2.1.

Quick Recap

Big O notation is a shortcut for describing how an algorithm's runtime scales as input grows — and counting your loops is usually all you need to estimate it.

3.4.2 Space Complexity

The Explanation

Space complexity measures how the *amount of memory* an algorithm uses changes as the size of the input grows. It helps us understand how efficiently an algorithm uses memory when handling large datasets.

Example: If an algorithm needs to store a list of numbers, its space complexity tells us how much memory will be required as the volume of numbers increases.

Interactive Stop-Point

Pause & Think: An algorithm that sorts a list "in place" (rearranging the existing list without creating a new one) uses very little extra memory. An algorithm that creates a brand-new copy of the list to sort it uses much more. Which one would you expect to have lower space complexity? Why might a developer sometimes choose the memory-heavier option anyway?

Quick Recap

Space complexity is time complexity's twin — instead of measuring *speed* as input grows, it measures *memory usage* as input grows.

3.5 Algorithm Design Techniques

The Hook (Story Mode)

Give five different chefs the same ingredients, and you'll likely get five different dishes — each shaped by a different overall strategy. Algorithm design is exactly like this. Faced with the same problem, computer scientists have discovered a handful of powerful, reusable *strategies* — general game plans that can be applied again and again to entirely different problems.

The Explanation

Algorithm design is a critical aspect of problem-solving in computer science. It involves creating systematic methods to solve problems efficiently and effectively. There are several well-known algorithm design techniques that help in developing robust algorithms for a wide variety of computational problems.

Quick Recap

Algorithm design techniques are reusable strategies — general game plans you can apply to many different problems, not just one-off tricks.

3.5.1 Divide and Conquer

The Hook (Story Mode)

Imagine merging two shuffled decks of cards. Instead of comparing every card against every other card at once, you could split each deck into smaller hands, sort each small hand individually, and then merge the sorted hands back together. That's **Divide and Conquer** — break the big problem into small, manageable pieces, solve each piece, then combine the results.

The Explanation

Divide and Conquer is a powerful algorithm design technique that works by breaking a large problem into smaller, more manageable parts. Each smaller part is solved independently, and then their solutions are combined to solve the original problem. This approach is especially effective for problems that can be divided into similar smaller sub-problems.

DIVIDE AND CONQUER – MERGE SORT PROCESS (Figure 3.6)

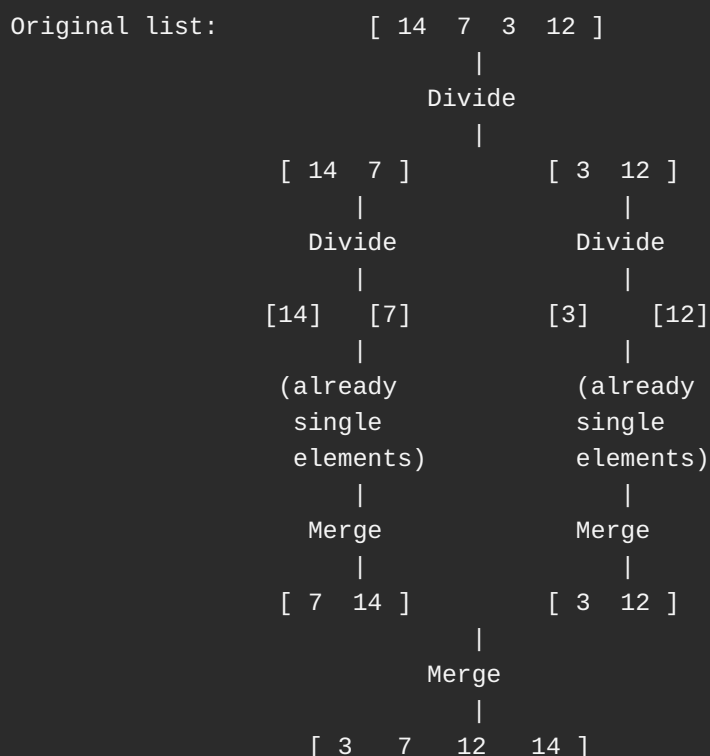


Figure 3.6: Merge Sort process using Divide and Conquer

Interactive Stop-Point

Pause & Think: Why might solving four small, simple problems (each with 2 elements) and merging them together be faster overall than trying to sort all 8 elements at once, comparing every possible pair?

Quick Recap

Divide and Conquer breaks a big, intimidating problem into smaller identical sub-problems — solving those is easy, and combining the results rebuilds the full solution.

3.5.2 Greedy Algorithms

The Hook (Story Mode)

Suppose you need to make change for 87 rupees using the fewest coins possible, and you have coins worth 50, 20, 10, 5, and 1. The natural instinct is to grab the biggest coin that still fits: one 50, then one 20, then one 10, then one 5, then two 1s. At every single step, you greedily grab the largest coin available — never looking back, never reconsidering earlier choices.

The Explanation

Greedy algorithms work by making a sequence of choices, each of which is **locally optimal** (the best choice *right now*), with the hope that these choices lead to a **globally optimal** solution overall. The greedy approach is often used when a problem has **optimal substructure** — meaning the best overall solution can be built from the best solutions to its smaller sub-problems.

Example — the Coin Change problem: Given coins of different denominations, make a specific amount using the fewest coins possible. The greedy algorithm chooses the largest denomination that doesn't exceed the remaining amount, subtracts it, and repeats.

Tidbit: Greedy algorithms are often faster and easier to implement than other techniques, but **they don't always guarantee the optimal solution** for every problem. Always analyze the problem carefully to make sure a greedy approach is actually appropriate.

Interactive Stop-Point

Pause & Think: You are filling a backpack with items that each have a different weight and value, and the backpack has a maximum weight limit. If your greedy strategy is "always pick the highest-value item first," why might this fail to give you the maximum possible total value once the backpack fills up?

Quick Recap

Greedy algorithms grab the best-looking option at each individual step — fast and simple, but they don't always add up to the *best possible* overall solution.

3.5.3 Dynamic Programming

The Hook (Story Mode)

In the 1950s, mathematician **Richard Bellman** was working at the RAND Corporation, doing mathematical optimization research for a government client who was suspicious of anything sounding too "mathematical." So Bellman deliberately coined a name that sounded impressive but harmless: "**Dynamic Programming**." The technique he was hiding behind that name turned out to be one of the most powerful ideas in all of computer science.

The Explanation

Dynamic Programming (DP) is an optimization technique that solves problems by breaking them into simpler **sub-problems**, and **storing the results** of those sub-problems so they never need to be recalculated. DP is especially useful for problems with **overlapping sub-problems** and **optimal substructure**.

Example: Calculating the **Fibonacci sequence**. Instead of recalculating the same Fibonacci numbers over and over through plain recursion, DP stores each result as it's computed, so future calculations can simply look up the answer instead of redoing the work — dramatically reducing the total number of calculations.

The Practical Walkthrough — Fibonacci with a Memoization Table

Goal: Calculate `fib(6)`, where `fib(n) = fib(n-1) + fib(n-2)`, and `fib(0) = 0`, `fib(1) = 1`.

1. **Create an empty memo table** to store already-computed results:

n	fib(n)
0	0
1	1

1. **Compute fib(2).** `fib(2) = fib(1) + fib(0) = 1 + 0 = 1`. Store it.

n	fib(n)
0	0
1	1
2	1

1. **Compute fib(3).** `fib(3) = fib(2) + fib(1) = 1 + 1 = 2`. Store it.

n	fib(n)
3	2

1. **Compute fib(4).** $\text{fib}(4) = \text{fib}(3) + \text{fib}(2) = 2 + 1 = 3$. Store it.
2. **Compute fib(5).** $\text{fib}(5) = \text{fib}(4) + \text{fib}(3) = 3 + 2 = 5$. Store it.
3. **Compute fib(6).** $\text{fib}(6) = \text{fib}(5) + \text{fib}(4) = 5 + 3 = 8$. Store it.

Final memo table:

n	0	1	2	3	4	5	6
fib(n)	0	1	1	2	3	5	8

What just happened? Instead of recalculating `fib(2)` and `fib(1)` over and over again inside a plain recursive call tree (which would happen without memoization), each value was calculated **exactly once** and simply looked up afterward — this is the entire power of Dynamic Programming.

Interactive Stop-Point

Pause & Think: Without a memo table, calculating `fib(6)` recursively would recompute `fib(2)` multiple separate times. Trace how many times a plain (non-memoized) recursive call would recalculate `fib(2)` while computing `fib(6)`, and compare that to the memoized version above.

Quick Recap

Dynamic Programming avoids redundant work by solving each sub-problem once, storing the answer, and reusing it — turning painfully slow recursive calculations into fast, efficient ones.

3.5.4 Backtracking

The Hook (Story Mode)

Imagine exploring a real maze. You walk down a corridor, and if it turns out to be a dead end, you don't give up — you simply walk back to the last junction and try a different corridor instead. That "try, fail, walk back, try again" pattern is exactly what **Backtracking** does inside an algorithm.

The Explanation

Backtracking is a method for building up a solution step by step. If a particular path doesn't lead to a solution, the algorithm simply goes back ("backtracks") and tries a different path. It's like trying out different routes on a map and turning back whenever you realize you're heading the wrong way. Backtracking is often used for problems where you must consider all possible options — such as puzzles or problems involving many different combinations.

Interactive Stop-Point

Pause & Think: Think of solving a maze on paper. At a junction with three possible paths, how would a backtracking algorithm decide which path to try first, and what would it do the moment it hits a dead end?

Quick Recap

Backtracking builds a solution one step at a time, and whenever a path leads to a dead end, it reverses course and tries a different option — instead of giving up entirely.

3.6 Commonly Used Algorithms

The Hook (Story Mode)

Every advanced piece of software you'll ever build rests on a small handful of foundational algorithms — the way every skyscraper rests on basic concrete and steel. In this final section, we're going to meet the algorithms every computer scientist knows by heart: how to **sort** data, how to **search** for something inside it, and how to **explore** connected networks of information.

The Explanation

Algorithms are essential tools in computer science, applied to a huge range of problems, from sorting data to searching for information in massive datasets. Some algorithms are foundational, serving as building blocks for far more complex operations. This section explores some of the most commonly used algorithms: sorting, searching, and graph traversal.

Quick Recap

Sorting, searching, and graph traversal algorithms are the foundational building blocks that nearly every larger piece of software depends on.

3.6.1 Sorting Algorithms

The Explanation

Sorting algorithms arrange data in a particular order — ascending or descending. Sorting is a fundamental operation that often serves as a *prerequisite* for other tasks, like searching and data analysis (as you'll see shortly, Binary Search only works on sorted data!).

Quick Recap

Sorting isn't just useful on its own — it's often the necessary first step that makes other algorithms, like Binary Search, possible.

3.6.1.1 Bubble Sort

The Hook (Story Mode)

Picture bubbles rising through water — smaller, lighter bubbles naturally drift upward past bigger, heavier ones. **Bubble Sort** works in a similar spirit: on each pass through the list, it compares neighbors and lets the "lighter" (smaller) values gradually bubble toward their correct position.

The Explanation

Bubble Sort is one of the simplest sorting algorithms. It works by repeatedly stepping through the list, comparing **adjacent elements**, and swapping them if they're in the wrong order. This process repeats until the list is fully sorted.

Process:

1. Start from the beginning of the list.
2. Compare each pair of adjacent elements.
3. Swap them if they're in the wrong order.
4. Continue the process until no more swaps are needed.

Complexity: Bubble Sort's time complexity is $O(n^2)$, making it inefficient for large datasets. However, it's easy to understand and implement, making it excellent for education and small datasets. This isn't a "bad" algorithm to have learned — it's the natural, intuitive starting point that helps you appreciate *why* faster algorithms like Merge Sort or Quick Sort were later invented.

The Practical Walkthrough — Bubble Sort on `[5, 2, 8, 1, 3]`

Pass 1:

Comparison	Action	List State
Start	—	[5, 2, 8, 1, 3]
Compare 5, 2	$5 > 2 \rightarrow$ swap	[2, 5, 8, 1, 3]
Compare 5, 8	$5 < 8 \rightarrow$ no swap	[2, 5, 8, 1, 3]
Compare 8, 1	$8 > 1 \rightarrow$ swap	[2, 5, 1, 8, 3]
Compare 8, 3	$8 > 3 \rightarrow$ swap	[2, 5, 1, 3, 8]

What just happened? After Pass 1, the largest element (8) has "bubbled" all the way to its correct final position at the end of the list.

Pass 2:

Comparison	Action	List State
Compare 2, 5	$2 < 5 \rightarrow$ no swap	[2, 5, 1, 3, 8]
Compare 5, 1	$5 > 1 \rightarrow$ swap	[2, 1, 5, 3, 8]
Compare 5, 3	$5 > 3 \rightarrow$ swap	[2, 1, 3, 5, 8]

Pass 3:

Comparison	Action	List State
Compare 2, 1	$2 > 1 \rightarrow$ swap	[1, 2, 3, 5, 8]
Compare 2, 3	$2 < 3 \rightarrow$ no swap	[1, 2, 3, 5, 8]

Pass 4: No swaps needed — the list [1, 2, 3, 5, 8] is fully sorted. Bubble Sort stops.

Tidbit: While Bubble Sort is easy to implement, consider using more efficient sorting algorithms like Quick Sort or Merge Sort for larger datasets to save time and resources.

Interactive Stop-Point

Pause & Think: Looking at the trace table above, how many total comparisons did it take to sort 5 elements? If the list had 10 elements instead of 5, would you expect roughly double the comparisons, or far more than double? Connect your answer to Bubble Sort's $O(n^2)$ complexity.

Quick Recap

Bubble Sort repeatedly swaps neighboring out-of-order elements, letting the largest values "bubble" to the end with each pass — simple to understand, but $O(n^2)$, so it's slow for large datasets.

3.6.1.2 Selection Sort

The Hook (Story Mode)

Imagine sorting a hand of playing cards by repeatedly scanning through the unsorted cards, picking out the single smallest one, and placing it at the front. That's **Selection Sort** — instead of comparing neighbors like Bubble Sort, it actively hunts for the minimum value each round.

The Explanation

Selection Sort works by selecting the **smallest** (or largest, depending on desired order) element from the *unsorted* part of the list, and swapping it with the first element of the unsorted part. This process repeats for the remaining unsorted portion.

Process:

1. Find the minimum element in the unsorted part of the list.
2. Swap it with the first unsorted element.
3. Move the boundary between sorted and unsorted sections forward by one.
4. Repeat for the remaining elements.

Complexity: Selection Sort's time complexity is also **$O(n^2)$** — like Bubble Sort, it's inefficient for large datasets, but straightforward to implement.

The Practical Walkthrough — Selection Sort on [29, 10, 14, 37, 13]

Round	Unsorted Portion	Minimum Found	Action	List State
Start	[29, 10, 14, 37, 13]	—	—	[29, 10, 14, 37, 13]
1	[29, 10, 14, 37, 13]	10	Swap 10 with 29	[10, 29, 14, 37, 13]
2	[29, 14, 37, 13]	13	Swap 13 with 29	[10, 13, 14, 37, 29]
3	[14, 37, 29]	14	Already first — no swap needed	[10, 13, 14, 37, 29]
4	[37, 29]	29	Swap 29 with 37	[10, 13, 14, 29, 37]
Done	[37] (last element, automatically in place)	—	—	[10, 13, 14, 29, 37]

What just happened? Each round, Selection Sort scans the entire remaining unsorted section to find the true minimum — this "scan the whole remainder" behavior is exactly why Selection Sort is also $O(n^2)$.

Interactive Stop-Point

Pause & Think: Bubble Sort swaps adjacent elements many times per pass, while Selection Sort finds the minimum first and swaps only once per round. Even though both are $O(n^2)$, which one do you think performs *fewer total swaps* on average? Why might that matter in a situation where swapping is an expensive operation?

Quick Recap

Selection Sort hunts for the smallest remaining element each round and places it directly into position — fewer swaps than Bubble Sort, but still $O(n^2)$ overall.

3.6.2 Search Algorithms

The Explanation

Search algorithms are designed to find specific elements — or a set of elements — within a dataset. They're critical for tasks like information retrieval, database queries, and decision-making processes.

Quick Recap

Whenever software needs to find something inside a pile of data, it's relying on a search algorithm — and which one it uses can make an enormous difference in speed.

3.6.2.1 Linear Search

The Hook (Story Mode)

Imagine flipping through a stack of unsorted photographs, one at a time, looking for a picture of your friend. You have no shortcut — you must check each photo individually until you find it (or reach the bottom of the stack). That's **Linear Search**.

The Explanation

Linear Search is the most straightforward method for finding an item in a list — you check each item one by one until you find what you're looking for.

Process:

1. **Start at the Beginning:** Look at the first item in the list.
2. **Check Each Item:** Compare the item you're looking for with the current item.
3. **Move to the Next:** If they don't match, move to the next item.
4. **Repeat:** Continue until you find the item, or reach the end of the list.

The Practical Walkthrough — Linear Search for "Islamabad"

List: [Karachi, Lahore, Islamabad, Faisalabad]

Step	Current Item	Match?	Action
1	Karachi	No	Move to next
2	Lahore	No	Move to next
3	Islamabad	Yes	Stop — found at position 3

What just happened? It took **3 comparisons** to find Islamabad. If it hadn't been in the list at all, Linear Search would have needed all **4** comparisons before concluding it wasn't there. This "check every item, worst case" behavior is why Linear Search has a time complexity of **$O(n)$** .

Interactive Stop-Point

Pause & Think: If a list has 1,000 items and the item you're searching for happens to be last in the list, how many comparisons will Linear Search need in the worst case?

Quick Recap

Linear Search checks items one by one, in order — simple and works on any list (sorted or not), but slow ($O(n)$) for large datasets.

3.6.2.2 Binary Search

The Hook (Story Mode)

Long before computers, telephone directory operators could locate any name among millions of alphabetically sorted entries in fewer than 20 steps — not by reading every name, but by opening the book to the middle, deciding "earlier or later in the alphabet?", and repeatedly cutting the remaining pages in half. That's **Binary Search** in action.

The Explanation

Binary Search is an efficient algorithm for finding an item in a **sorted** list. It works by repeatedly dividing the search interval in half, discarding the half where the item cannot be, until the item is found or the interval is empty.

Process:

1. Start with the **middle** element of the sorted list.
2. If the middle element is the target, return its position.
3. If the target is **smaller** than the middle element, repeat the search on the **left half**.
4. If the target is **larger**, repeat the search on the **right half**.

Complexity: Binary Search's time complexity is **$O(\log n)$** — dramatically faster than Linear Search for large datasets.

Did You Know? Binary Search only works on **sorted** lists. If your data isn't sorted, you'll need to sort it first (e.g., with Merge Sort) before applying Binary Search!

The Practical Walkthrough — Binary Search for 78 in

[21, 34, 43, 57, 66, 78]

Step	Current Range	Middle Element	Comparison	Action
1	[21, 34, 43, 57, 66, 78]	43 (index 2)	78 > 43	Search right half
2	[57, 66, 78]	66 (index 4)	78 > 66	Search right half
3	[78]	78 (index 5)	78 == 78	Found!

What just happened? Binary Search found the target in just **3 steps**, compared to what could have taken up to 6 steps with Linear Search — and this gap grows dramatically as the list gets bigger. On a sorted list of 1,000,000 items, Binary Search needs at most about 20 steps, while Linear Search could need up to 1,000,000.

Figure 3.6: Binary Search Process — halving the search interval at each step

Interactive Stop-Point

Grab a Partner: Compare the guess-count results from your earlier Binary Search vs. Linear Search "number guessing" activity in section 3.4.1.1. Now formally connect that experience to this walkthrough: why does cutting a list in half repeatedly beat checking every item one at a time, especially as the list size grows into the thousands or millions?

Quick Recap

Linear search looks at items one by one, while binary search repeatedly cuts the search space in half — making it dramatically faster ($O(\log n)$) on large, sorted datasets.

3.6.3 Graph Algorithms

The Hook (Story Mode)

In 1736, mathematician **Leonhard Euler** was asked whether it was possible to walk through the city of Königsberg crossing each of its seven bridges exactly once, returning to the starting point. Euler proved it was impossible — and in doing so, he represented the city as a set of connected points and lines, inventing an entirely new branch of mathematics: **graph theory**. Two centuries later, that same idea would power GPS navigation, social networks, and the internet itself.

The Explanation

Graph algorithms explore and analyze **graphs** — data structures made up of **nodes** (also called vertices) connected by **edges**. These algorithms are essential for network analysis, route planning, and social network analysis.

Quick Recap

A graph is simply a collection of points (nodes) connected by lines (edges) — and graph algorithms are the tools we use to explore, search, and analyze those connections.

3.6.3.1 Breadth-First Search (BFS)

The Hook (Story Mode)

Imagine dropping a stone into still water. The ripple spreads outward evenly, in expanding rings, reaching everything at one distance before moving on to the next. That's exactly how **Breadth-First Search (BFS)** explores a graph — level by level, like ripples in water.

The Explanation

Breadth-First Search (BFS) is a graph traversal algorithm that explores all the nodes of a graph **level by level**, starting from a given node (often called the root). It uses a **queue** (first-in, first-out) to keep track of which nodes still need to be explored.

Process:

1. Start from the root node and **enqueue** it.
2. **Dequeue** a node, process it, and enqueue all of its unvisited neighbors.
3. Repeat until the queue is empty.

Complexity: BFS has a time complexity of $O(V + E)$, where V is the number of vertices (nodes) and E is the number of edges — making it efficient for exploring even large graphs.

Example: In a social network graph, where each node represents a person and edges represent friendships, BFS can find the shortest path between two people — for example, finding the "degree of separation" between two users.

The Practical Walkthrough — BFS on a Graph Starting at Node 0

Graph structure (Figure 3.7): Node 0 connects to nodes 1, 2, 3. Node 1 connects to nodes 4, 5. Node 2 connects to node 6. Node 3 connects to node 7.

Step	Queue Before	Node Processed	Neighbors Enqueued	Queue After
1	[0]	0	1, 2, 3	[1, 2, 3]
2	[1, 2, 3]	1	4, 5	[2, 3, 4, 5]
3	[2, 3, 4, 5]	2	6	[3, 4, 5, 6]
4	[3, 4, 5, 6]	3	7	[4, 5, 6, 7]
5	[4, 5, 6, 7]	4	(none unvisited)	[5, 6, 7]
6	[5, 6, 7]	5	(none unvisited)	[6, 7]
7	[6, 7]	6	(none unvisited)	[7]
8	[7]	7	(none unvisited)	[] (empty — stop)

BFS Output order: 0, 1, 2, 3, 4, 5, 6, 7

What just happened? Notice how BFS fully processes **Layer 0** (node 0), then all of **Layer 1** (nodes 1, 2, 3), then all of **Layer 2** (nodes 4, 5, 6, 7) — exactly like ripples expanding outward one ring at a time.

Interactive Stop-Point

Pause & Think: If you wanted to find the *shortest* number of friendship connections between two people in a social network, would BFS's "layer by layer" exploration naturally guarantee you find the shortest path first? Why?

Quick Recap

BFS explores a graph level by level using a queue — perfect for finding the shortest path in an unweighted graph, since it processes closer nodes before farther ones.

3.6.3.2 Depth-First Search (DFS)

The Hook (Story Mode)

Now imagine exploring that same maze differently: instead of checking every nearby hallway exit first, you commit to a single corridor and follow it as far as it goes — turning back only when you hit a dead end. That's **Depth-First Search (DFS)** — it runs deep down one branch before ever backtracking to try another.

The Explanation

Depth-First Search (DFS) is a graph traversal algorithm that explores as far down a branch as possible before backtracking to explore other branches. It uses a **stack** (last-in, first-out) to manage which nodes are still waiting to be explored.

Process:

1. Start from the root node and **push** it onto the stack.
2. **Pop** a node, process it, and push all of its unvisited neighbors onto the stack.
3. Repeat until the stack is empty.

Complexity: DFS also has a time complexity of $O(V + E)$, similar to BFS. However, DFS is more **memory-efficient for deep graphs**, while BFS is more suited for **shallow, wide graphs**.

Example: DFS is often used for solving puzzles like mazes — the algorithm explores one possible path all the way to the end, and if it hits a dead end, it backtracks and tries another path.

The Practical Walkthrough — DFS on the Same Graph Starting at Node 0

Graph structure: Node 0 connects to nodes 1, 2, 3. Node 1 connects to nodes 4, 5. Node 2 connects to node 6. Node 3 connects to node 7.

Step	Stack Before	Node Processed (popped)	Neighbors Pushed	Stack After
1	[0]	0	1, 2, 3	[1, 2, 3]
2	[1, 2, 3]	3	7	[1, 2, 7]
3	[1, 2, 7]	7	(none unvisited)	[1, 2]
4	[1, 2]	2	6	[1, 6]
5	[1, 6]	6	(none unvisited)	[1]
6	[1]	1	4, 5	[4, 5]
7	[4, 5]	5	(none unvisited)	[4]
8	[4]	4	(none unvisited)	[] (empty — stop)

DFS Output order (using this stack ordering): 0, 3, 7, 2, 6, 1, 5, 4

(Note: the exact DFS output order can vary slightly depending on the order neighbors are pushed onto the stack — some textbook versions instead push neighbors in ascending order and pop from the front, producing outputs like 0, 1, 4, 5, 2, 6, 3, 7 as shown in Figure 3.7. Either version is valid DFS — the defining feature is always going as deep as possible before backtracking.)

What just happened? Unlike BFS's neat, level-by-level spread, DFS plunges straight down one branch (0 → 3 → 7) before ever circling back to explore node 2's branch — exactly like committing to one long hallway in a maze before backtracking.

Figure 3.7: Comparison of BFS and DFS traversal patterns

Interactive Stop-Point

Pause & Think: For solving a maze where you just need to find *any* valid path to the exit (not necessarily the shortest one), would DFS or BFS likely use less memory while searching? Why does "going deep first" sometimes save memory compared to "exploring every direction at once"?

Quick Recap

DFS dives as deep as possible down one path before backtracking, using a stack — memory-efficient for deep graphs, though it doesn't guarantee the shortest path the way BFS does.

Chapter Summary — The Big Picture

```
ALGORITHMIC THINKING, END TO END

[ Characterize the Problem: well-defined? decision, search, optimization, or
counting? ]
      |
      [ Is it even Solvable? (Halting Problem shows some things
aren't) ]
      |
      [ How Complex is it? Tractable (P) or Intractable (NP-hard/NP-
complete)? ]
      |
      [ Analyze Candidate Algorithms: Time Complexity (Big O) & Space
Complexity ]
      |
      [ Choose a Design Technique: Divide & Conquer, Greedy, Dynamic Programming,
Backtracking ]
      |
      [ Apply a Concrete Algorithm: Bubble/Selection Sort, Linear/Binary Search,
BFS/DFS ]
```

Every idea in this unit connects to the next. You characterize a problem, confirm it's solvable, judge how hard it really is, measure candidate algorithms mathematically, choose a design strategy, and finally apply a concrete, well-known algorithm. This is the exact mental checklist professional software engineers run through — often without even realizing it — every time they face a new problem.

You now think like an algorithmic problem solver. The next step is practicing these ideas on new problems until this checklist becomes second nature.

Exercise

Multiple Choice Questions

1. The characteristic of a well-defined problem is:
 - a) Ambiguous goals and unclear requirements
 - b) Vague processes and inputs
 - c) Clear goals, inputs, processes, and outputs
 - d) Undefined solutions
2. Complexity class representing problems solvable efficiently by a deterministic algorithm:
 - a) NP
 - b) NP-hard
 - c) NP-complete
 - d) P
3. The statement that applies to unsolvable problems:
 - a) They can be solved in polynomial time
 - b) They cannot be solved by any algorithm
 - c) They are always in NP class
 - d) They require exponential time to solve
4. The meaning of NP in computational complexity is:
 - a) Non-deterministic Polynomial time
 - b) Negative Polynomial time
 - c) Non-trivial Polynomial time
 - d) Numerical Polynomial time
5. Search algorithm more efficient for large datasets:
 - a) Bubble Sort
 - b) Merge Sort
 - c) Selection Sort
 - d) Quick Sort
6. A scenario where Dynamic Programming proves most useful:
 - a) Problems without overlapping subproblems
 - b) Problems solved by making local choices
 - c) Problems with overlapping subproblems and optimal substructure
 - d) Problems divided into independent subproblems

7. An algorithm that sorts data by stepping through the list and swapping adjacent elements if needed is:
- a) Selection Sort
 - b) Quick Sort
 - c) Bubble Sort
 - d) Merge Sort
8. Time complexity of Depth-First Search (DFS) in a graph is:
- a) $O(n \log n)$
 - b) $O(V)$
 - c) $O(V + E)$
 - d) $O(n)$
9. Best description of time complexity:
- a) Amount of memory an algorithm needs
 - b) Time taken as a function of input size
 - c) Efficiency as input size grows
 - d) Upper bound of space requirements
10. An algorithm with a time complexity of $O(n \log n)$:
- a) Bubble Sort
 - b) Binary Search
 - c) Merge Sort
 - d) Insertion Sort

Short Questions

1. Differentiate between well-defined and ill-defined problems within the realm of computational problem-solving.
2. Outline the main steps involved in the Generate-and-Test method.
3. Compare tractable and intractable problems in the context of computational complexity.
4. Summarize the key idea behind Greedy Algorithms.
5. Discuss the advantages of using Dynamic Programming.
6. Compare the advantages of Breadth-First Search (BFS) with Depth-First Search (DFS) in graph traversal.
7. Explain the importance of breaking down a problem into smaller components in algorithmic thinking.
8. Identify the key factors used to evaluate the performance of an algorithm.

Long Questions

1. Provide a detailed explanation of why the Halting Problem is considered unsolvable and its implications in computer science.
 2. Discuss the characteristics of search problems and compare the efficiency of Linear Search and Binary Search algorithms.
 3. Discuss the nature of optimization problems and provide examples of their applications in real-world scenarios.
 4. Explain the process and time complexity of the Bubble Sort algorithm. Compare it with another sorting algorithm of your choice in terms of efficiency.
 5. Discuss the differences between time complexity and space complexity. How do they impact the choice of an algorithm for a specific problem?
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Key Vocabulary Glossary

Term	Plain-English Definition
Computational Problem	A challenge solvable through an algorithm's input → process → output.
Algorithm	A precise, step-by-step recipe a computer can follow.
Decision Problem	A problem whose answer is simply yes or no.
Search Problem	A problem that asks you to find something meeting given criteria.
Optimization Problem	A problem that asks for the <i>best</i> solution among many.
Counting Problem	A problem that asks how many ways something can be done.
Well-defined Problem	A problem with clear goal, input, process, and output.
Ill-defined Problem	A problem with vague or ambiguous goals or requirements.
Generate-and-Test	A brute-force strategy: generate candidate solutions, test each one.
Solvable Problem	A problem an algorithm can solve in finite time.
Unsolvable Problem	A problem no algorithm can solve for every possible case (e.g., the Halting Problem).
Tractable Problem	A problem solvable in polynomial time (efficiently solvable).
Intractable Problem	A problem requiring super-polynomial (often exponential) time.
Class P	Problems solvable efficiently (in polynomial time).
Class NP	Problems whose proposed solutions can be verified quickly.
NP-Hard	Problems at least as hard as the hardest problems in NP.
NP-Complete	Problems that are both in NP and as hard as any problem in NP.
Time Complexity	How an algorithm's runtime grows as input size grows.
Space Complexity	How an algorithm's memory usage grows as input size grows.
Big O Notation	Mathematical shorthand describing an algorithm's worst-case growth rate.
O(1)	Constant time — runtime doesn't change with input size.
O(log n)	Logarithmic time — runtime grows very slowly (e.g., Binary Search).
O(n)	Linear time — runtime grows directly with input size.
O(n²)	Quadratic time — runtime grows with the square of input size.

Term	Plain-English Definition
Divide and Conquer	Breaking a problem into smaller sub-problems, solving them, and combining results.
Greedy Algorithm	Making the locally best choice at each step, hoping for a globally optimal result.
Dynamic Programming	Solving overlapping sub-problems once and storing (memoizing) results for reuse.
Backtracking	Building a solution step by step, reversing course whenever a path fails.
Bubble Sort	Repeatedly swapping adjacent out-of-order elements until sorted; $O(n^2)$.
Selection Sort	Repeatedly selecting the minimum remaining element and placing it in position; $O(n^2)$.
Linear Search	Checking each item one by one until found; $O(n)$.
Binary Search	Repeatedly halving a sorted list's search interval; $O(\log n)$.
Graph	A structure of nodes (vertices) connected by edges.
BFS (Breadth-First Search)	Exploring a graph level by level using a queue; $O(V + E)$.
DFS (Depth-First Search)	Exploring a graph as deep as possible before backtracking, using a stack; $O(V + E)$.

End of Unit 3 — Algorithms and Problem Solving.